CSA5108 Cryptography and Network security

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Exp 15 : Automatic Letter-Frequency Attack on an Additive Cipher
Code:

```
#include <stdio.h>
#include <string.h>
#include <ctype.h>

int main(void) {
   char cipher[2000], plain[2000];
   int top, freq[26]={0}, shifts[26];

   printf("Enter ciphertext: ");
   fgets(cipher,sizeof(cipher),stdin);

   printf("How many possible plaintexts? ");
   scanf("%d",&top);

   if(top<1) top=1;
   if(top>26) top=26;

   for(int i=0;cipher[i];i++)
      if(isalpha((unsigned char)cipher[i]))
         freq[tolower((unsigned char)cipher[i])-'a']++;

   for(int i=0;i<26;i++) shifts[i]=i;

   for(int i=0;i<26;i++)
      for(int j=i+1;j<26;j++)
         if(freq[shifts[j]]>freq[shifts[i]]) {
            int t=shifts[i];
            shifts[i]=shifts[j];
            shifts[j]=t;
         }
```

```
    for(int s=0;s<top;s++) {
        int shift=(shifts[s]-'e'+'a'+26)%26;

        printf("\nCandidate %d (shift %d):\n",s+1,shift);

        for(int i=0;cipher[i];i++) {
            if(isupper((unsigned char)cipher[i]))
                plain[i]=(cipher[i]-'A'-shift+26)%26+'A';
            else if(islower((unsigned char)cipher[i]))
                plain[i]=(cipher[i]-'a'-shift+26)%26+'a';
            else plain[i]=cipher[i];
        }
        plain[strlen(cipher)]='\0';
        printf("%s",plain);
    }

    return 0;
}
```

